Stage-Localized Linearization and Equal-Power DPD Evaluation of a 26 GHz Heterodyne Transmitter

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***Abstract*—This paper presents the behavioral linearization of a cascaded heterodyne transmitter using memory-polynomial (MP) digital predistortion (DPD). Stage-wise sweeps verified approximately linear intermediate-frequency (IF) and observation paths while the integrated upconverter was operated in mild compression. Three regularized MP models with 6–36 coefficients were compared; results demonstrate that a low-complexity structure achieves EVM and ACPR values comparable to those of a high-complexity model. To separate linearization from power back-off, DPD was evaluated against an independent no-DPD sweep at equal received and detector-measured power. At the nominal operating point, DPD reduced EVM from 6.47% to 2.65% and improved ACPR from -26.57 to -30.88 dBc. Across 20 paired runs referenced to equal detector power, mean improvements were 3.24 percentage points in EVM and 1.36 dB in ACPR.**

***Index Terms*—Digital predistortion (DPD), behavioral linearization, millimeter-wave (mmWave), memory polynomial, software-defined radio (SDR), 5G NR, equal-power evaluation.**

I. Introduction

Fifth-generation New Radio (5G NR) systems operating in Frequency Range 2 (FR2) employ wideband orthogonal frequency-division multiplexing (OFDM) waveforms and high-order modulation to achieve high spectral efficiency [1], [2]. For instance, under 3GPP TS 38.104, a 256-QAM downlink signal imposes a stringent error vector magnitude (EVM) requirement of 3.5% and adjacent channel leakage ratio (ACLR) limits in the order of -30 dBc. These tight margins render uncompensated amplifier nonlinearities unacceptable, making linearization a design requirement rather than an optional enhancement.

Digital predistortion (DPD) compensates these effects by applying an approximate inverse nonlinear response in complex baseband. Memory-polynomial (MP) and generalized memory-polynomial (GMP) structures are widely used with indirect learning architectures because they capture nonlinear memory effects with manageable implementation cost [3], [4]. DPD has been widely demonstrated on software-defined radio (SDR) platforms, predominantly in the sub-6 GHz range. However, experimental DPD evaluation on low-cost commercial-off-the-shelf (COTS) millimeter-wave (mmWave) platforms presents distinct methodological difficulties that are less prevalent in sub-6 GHz systems.

A primary challenge arises when the integrated feedback receiver enters compression, causing the learned model to inadvertently capture both transmitter and receiver nonlinearities [5]. In addition, increasing memory depth and nonlinear order raise the coefficient count and least-squares numerical sensitivity without guaranteeing proportional accuracy [6]. Fixed-step iterative learning may also degrade after the first updates, as measurement noise and synchronization errors accumulate in the loop. To mitigate these extraction issues, the proposed methodology employs a predefined offline EVM–ACPR selection rule and does not claim autonomous online convergence.

Beyond the extraction phase, a critical evaluation challenge arises because predistortion increases waveform peak excursion. Digital normalization to preserve headroom may then reduce average RF power. Consequently, comparisons performed at the same programmable gain can erroneously attribute part of the linearization benefit to power back-off. In our measurements, this effect is not negligible: the DPD waveform reduced the average observed RF power by more than 2 dB, confirming that equal-power referencing is strictly required to avoid inflated performance claims.

Rather than proposing a new algorithm, the present contribution is experimental and methodological. Because reported

linearization gains often rely on single captures without equal-power referencing, the goal is to quantify the true linearization benefit on a low-cost heterodyne chain under repeatable, power-referenced conditions. The methodology linearizes the complete cascaded 5 GHz IF-to-26 GHz RF path from its end-to-end complex response. The main contributions are:

1) a stage-wise procedure that localizes the dominant nonlinear contribution and verifies approximately linear IF and observation paths;
2) a comparative evaluation of three MP structures spanning low, medium, and high complexity;
3) an equal-power assessment using both SDR-received power and an independent RMS detector;
4) local operating-range verification and statistical repeatability over 20 paired hardware runs generated with different 5G waveform seeds.

II. Experimental Testbed and Nonlinearity Localization

A. Experimental Testbed

A MATLAB-generated 20 MHz 5G NR downlink waveform with 30 kHz subcarrier spacing and full-band 256-QAM PDSCH is transmitted by an ADALM-Pluto SDR at 5 GHz IF. Samples are peak-normalized to 0.8, yielding approximately 11.3 dB PAPR.

The 5 GHz IF signal is amplified by a two-stage IF block and applied to an ADMV1018 up/downconverter [7]. The device operates as an upconverter in the transmit path, and its 26 GHz RF output is filtered by a bandpass filter with approximately 3 dB insertion loss. A 30 dB attenuator then connects the transmitter output to the same ADMV1018 device, which is reconfigured as a downconverter for the observation path. The recovered 5 GHz IF is captured by the Pluto receiver and used for synchronization, DPD training, and waveform-quality evaluation.

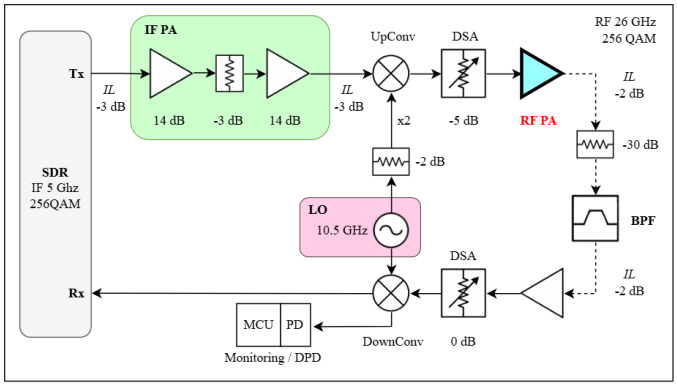
An LTC5596 linear-in-dB RMS detector interfaced to an STM32 microcontroller provides an independent relative-power measurement along the observation path [8]. The final characterization and DPD measurements use 5 dB of transmit-side DSA attenuation in the integrated upconverter. The complete loopback block diagram and the experimental setup photograph are shown in Fig. 1(a) and Fig. 1(b), respectively.

B. Stage-by-Stage Compression Characterization

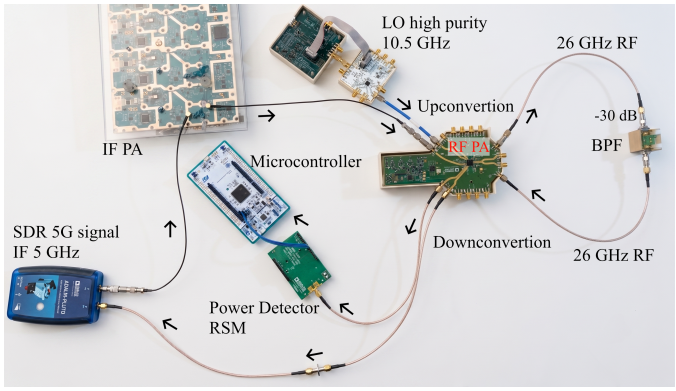
Independent power sweeps were performed for the Pluto transmitter, the complete IF amplification block, the upconverter, and the downconversion observation path. For each stage, the small-signal gain reference was obtained from the approximately linear portion of the sweep. Gain compression at operating point i was defined as

$$C_i = G_{\mathrm{SS}} - G_i, \tag{1}$$

where G_{SS} is the small-signal gain and G_i is the measured or reconstructed gain. The 1-dB compression point was estimated by interpolation between adjacent samples surrounding $C_i = 1$ dB.



(a)



(b)

Fig. 1. Conducted 26 GHz experimental platform: (a) Block diagram where a 5 GHz IF signal is amplified, upconverted, filtered, attenuated by 30 dB, downconverted, and acquired by the SDR observation receiver. (b) Photograph of the assembled laboratory setup. The STM32-interfaced RMS detector provides an independent relative-power reference.

TABLE I
Stage-wise compression at the selected operating point

Stage	G_{TX} **at P1dB (dB)**	C **at** -22.25 **dB (dB)**
Pluto TX	Not reached	≈ 0
IF block	Not reached	≈ 0
Upconverter	-19.78	0.50
Downconverter	Not reached	≈ 0

Both the Pluto output and the IF block remained approximately linear near the selected operating point. With the upconverter DSA fixed at 5 dB, the block-level response indicated that compression was dominated by the final upconversion stage. Consequently, the nonlinear device under test was treated as the integrated upconversion stage rather than an isolated discrete PA. The 30 dB loopback attenuation further ensured that the downconverter remained linear while the upconverter compressed. Fig. 2 and Table I summarize the results. At $G_{\mathrm{TX}} = -22.25$ dB, the upconverter exhibited approximately 0.5 dB compression, whereas the IF and downconversion paths remained within the measurement dispersion around their linear responses.

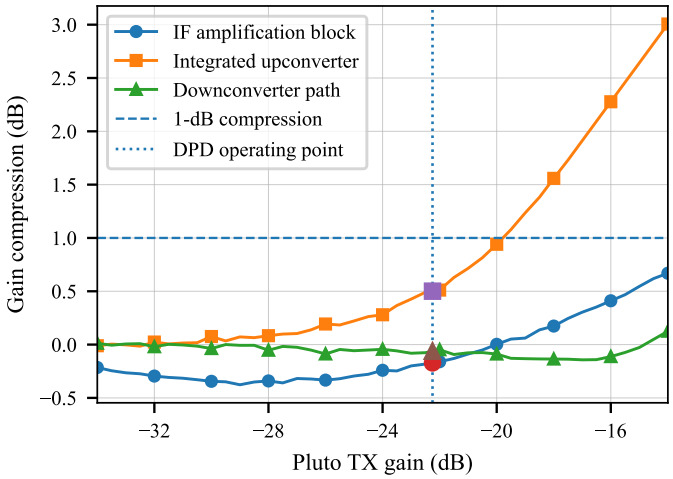


Fig. 2. Stage-wise gain compression versus Pluto transmit gain. The horizontal line denotes the 1-dB compression criterion, and the vertical marker identifies the selected DPD operating point, $G_{\mathrm{TX}} = -22.25$ dB. The IF and downconversion paths remain near zero compression at this point, whereas the integrated upconverter exhibits approximately 0.5 dB compression.

C. DPD Operating-Point Selection

An independent no-DPD full-chain sweep was acquired with the same 5-dB transmit DSA, 30-dB loopback attenuation, and fixed receiver configuration. For each transmit-gain point, waveform-quality metrics (synchronization, EVM, ACPR, aligned NMSE), power levels (received and detector-measured), and receiver peak amplitude were recorded. Captures were accepted only when the synchronization metric exceeded 0.95, aligned NMSE was below -10 dB, and the received peak remained below -3 dBFS.

The selected operating point, $G_{\mathrm{TX}} = -22.25$ dB, was intentionally located below the measured 1-dB compression point. This operating point provided measurable nonlinear degradation while preserving waveform-peak headroom for predistortion, maintaining synchronization, and avoiding compression in the observation path. This work therefore addresses behavioral linearization under moderate compression rather than operation in deep saturation. The same no-DPD sweep served as the reference for the equal-power comparisons.

III. Memory-Polynomial DPD Methodology

A. Memory-Polynomial Structure

For a complex input sequence $x[n]$, the predistorted waveform is

$$z[n] = \sum_{m=0}^{M} \sum_{k=0}^{K-1} c_{m,k} x[n-m] \, |x[n-m]|^{2k}, \tag{2}$$

where M denotes the maximum memory lag, so that $M+1$ delayed taps are used, K is the number of nonlinear basis branches, and $c_{m,k}$ are complex coefficients. The implemented basis therefore contains odd orders $1, 3, \ldots, 2K-1$, and the total coefficient count is

$$N_c = (M+1)K. \tag{3}$$

Three evaluated structures spanned low (M1K3), medium (M3K5), and high complexity (M5K6), with total coefficients (N_c) ranging from 6 to 36.

B. Regularized Iterative Indirect Learning

At each hardware iteration, the aligned transmitted and received waveforms formed a postdistorter identification problem

$$\mathbf{y} = \mathbf{X}\mathbf{c} + \boldsymbol{\varepsilon}, \tag{4}$$

where $\mathbf{X}$ is the MP regression matrix. To mitigate ill-conditioning from correlated basis functions and observation noise, coefficients were extracted via Tikhonov-regularized least squares [9], [10]:

$$\hat{\mathbf{c}} = \left(\mathbf{X}^H\mathbf{X} + \lambda\mathbf{I}\right)^{-1}\mathbf{X}^H\mathbf{y}. \tag{5}$$

The regularization parameter λ was conservatively set to 10^{-2} relative to the regression-matrix energy, providing moderate regularization while obviating a full sensitivity sweep.

To reduce abrupt capture-to-capture coefficient changes, the update was blended as

$$\mathbf{c}^{(i+1)} = (1-\mu)\mathbf{c}^{(i)} + \mu\hat{\mathbf{c}}^{(i)}, \tag{6}$$

with an empirically fixed step size $\mu = 0.12$.

The waveform-domain linearization error was quantified using an end-to-end aligned normalized mean-square error (NMSE),

$$\mathrm{NMSE}_{\mathrm{E2E,dB}} = 10\log_{10}\left(\frac{\|\mathbf{x}_{\mathrm{des}} - \alpha\mathbf{y}_{\mathrm{al}}\|_2^2}{\|\mathbf{x}_{\mathrm{des}}\|_2^2}\right), \tag{7}$$

where $\mathbf{x}_{\mathrm{des}}$ is the same original desired waveform for both the no-DPD and DPD cases, $\mathbf{y}_{\mathrm{al}}$ is the time- and frequency-aligned received waveform, and α compensates residual linear gain and phase. Both numerator and denominator in (7) are squared norms, i.e., waveform energies. A more negative NMSE indicates lower normalized waveform error. During acquisition, an input-referenced NMSE was also used as a capture-quality indicator. Since the DPD acquisition originally used the predistorted drive waveform as that internal reference, the reported end-to-end NMSE values were recomputed offline using the common desired waveform $\mathbf{x}_{\mathrm{des}}$.

Each structure was trained for six iterations. EVM, ACPR, PAPR, received power, coefficient norm, coefficient-update norm, and a condition indicator were recorded at every iteration. No independent validation waveform or validation-based early stopping was used. Several models reached minimum EVM within the first updates and subsequently degraded. The retained iteration was therefore selected offline using a predefined EVM–ACPR plateau criterion, subject to capture-quality and power-safety checks. The 0.1-dB ACPR tolerance used in the plateau rule was selected as a practical narrow margin, intended to avoid retaining an iteration with a marginal ACPR advantage but substantially degraded EVM. This procedure is an offline model-selection rule and should not be interpreted as an autonomous stopping mechanism.

C. Model and Iteration Selection

One representative structure per complexity class was selected. For each model, iterations within 0.1 dB of its best ACPR were identified, and the lowest-EVM iteration among them was retained.

The three-model comparison and the local-range and repeatability measurements were performed in separate hardware sessions. Their M5K6 coefficient sets are therefore treated as independent realizations of the same model structure. The detailed PSD, AM/AM, AM/PM, and end-to-end NMSE results correspond to the representative M5K6 realization retained in the three-model comparison, whereas the local-range and repeatability measurements used a frozen M5K6 realization obtained in the earlier session.

M5K6 is retained for detailed behavioral plots as the maximum-complexity MP reference, not because it is strictly superior to M1K3 in the representative run. This allows inspection of the richest AM/AM and AM/PM correction action within the tested MP class, while M1K3 is identified as the preferred low-complexity implementation.

D. Equal-Power Evaluation Protocol

Predistortion expands waveform peaks. To preserve digital headroom, the DPD waveform is normalized before transmission, which may reduce its average observed RF power. A comparison performed only at the same programmable transmit gain can therefore overestimate the predistortion benefit.

Each DPD capture was compared with the independent no-DPD sweep using two power references: average digital power measured by the Pluto receiver and power measured by the STM32-interfaced detector. For DPD power P_{DPD}, the corresponding no-DPD metric was obtained through interpolation:

$$\mathcal{M}_{\mathrm{NoDPD,eq}} = \mathrm{interp}(P_{\mathrm{NoDPD}}, \mathcal{M}_{\mathrm{NoDPD}}, P_{\mathrm{DPD}}), \qquad (8)$$

where $\mathcal{M}$ denotes EVM or ACPR. End-to-end NMSE was not power-interpolated because the required common-reference waveform captures were not available over the complete power sweep. Accordingly, NMSE comparisons should be read as nominal-capture waveform-error improvements, not as equal-power metrics. The net EVM and ACPR improvements were therefore computed relative to the no-DPD waveform at equivalent power.

IV. Experimental Results

A. Model-Complexity Comparison

All three structures were evaluated at $G_{\mathrm{TX}} = -22.25$ dB and $G_{\mathrm{RX}} = 8$ dB. Table II reports one representative model from each complexity class, while Fig. 4 shows their iteration-wise EVM evolution. Evaluation of these three models confirmed that adding coefficients did not yield monotonic improvement.

The aligned waveform NMSE was not exported for every model and iteration. It was therefore not included in Table II, avoiding a mixture of waveform-domain and symbol-domain NMSE definitions.

TABLE II
Representative MP model performance

Class	Model	N_c	Iter.	EVM (%)	ACPR (dBc)
Low	M1K3	6	1	2.83	−31.11
Medium	M3K5	20	4	3.00	−30.99
High	M5K6	36	4	2.86	−30.97

TABLE III
DPD improvement at the nominal operating point

Reference condition	ΔEVM (pp)	ΔACPR (dB)
Same TX gain	3.82	4.31
Same RX power	3.06	1.01
Same detector power	3.50	1.45

Most models reached minimum EVM in the first updates and then degraded with only small ACPR changes, confirming that the retained iterations were offline selections. The six-coefficient M1K3 model reached 2.83% EVM and −31.11 dBc ACPR, comparable to the selected M5K6 realization with substantially fewer coefficients. As noted in Section III, M1K3 is the preferred low-complexity implementation; M5K6 serves as the maximum-complexity behavioral reference.

B. Equal-Power and Local Operating-Range Verification

At the nominal operating point, the frozen predistorter reduced EVM from 6.47% to 2.65% and improved ACPR from −26.57 to −30.88 dBc. Since the DPD waveform necessarily reduced both received and detector-measured power, Table III contrasts same-gain and equal-power metrics. Equal-power referencing isolates the behavioral linearization contribution, whereas same-gain comparisons conflate it with DPD-induced power back-off.

To verify stability beyond this nominal point, a local operating-range sweep was conducted using the frozen M5K6 coefficient set. Sweeping the transmit gain from −22.25 down to −25 dB in 0.25-dB increments (Fig. 6), the corrected EVM remained between 2.16% and 2.90%. Simultaneous EVM and ACPR improvement at equivalent detector power was observed down to approximately −23.50 dB, confirming the local validity of the static coefficients on the back-off side.

C. Spectral and Behavioral Linearization

The normalized spectra in Fig. 5 show reduced adjacent-channel regrowth after predistortion, consistent with the measured ACPR improvement.

The measured AM/AM response in Fig. 3(a) shows the compression of the uncorrected cascaded path, the amplitude expansion introduced by the predistorter, and the resulting linearized output. The AM/PM response in Fig. 3(b) shows the compensating amplitude-dependent phase action and the residual phase distortion. These plots provide qualitative behavioral evidence of the predistortion mechanism; quantitative in-band improvement is reported through EVM and NMSE.

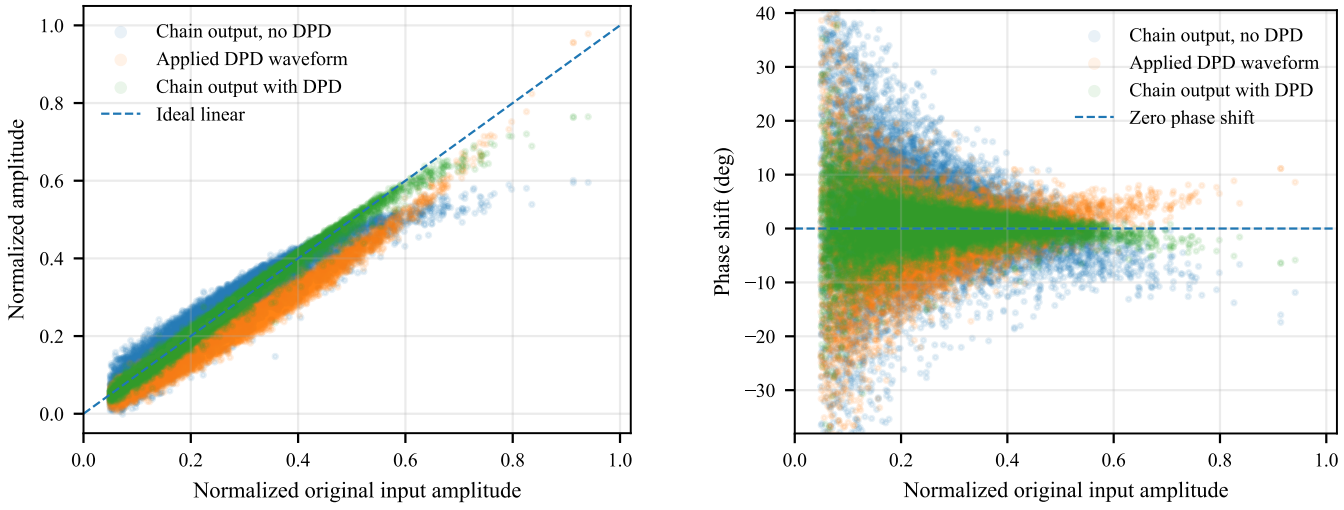


Fig. 3. Measured behavioral response of the cascaded transmitter: (a) AM/AM response and DPD amplitude action; (b) AM/PM response and compensating phase action.

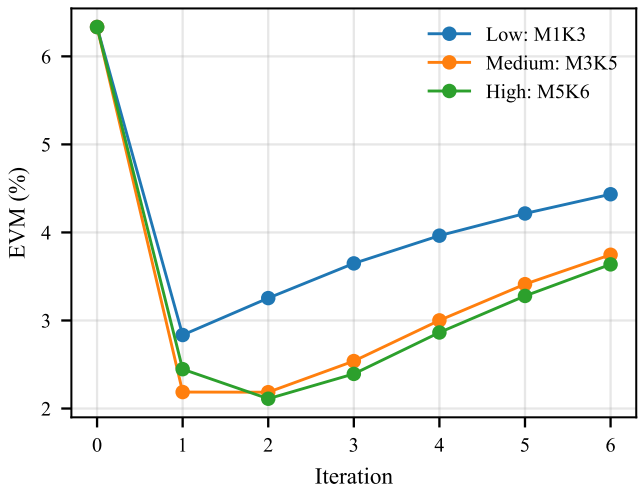


Fig. 4. Iteration-wise EVM evolution for representative low-, medium-, and high-complexity MP structures. All three candidates were evaluated; one model per class is shown for clarity.

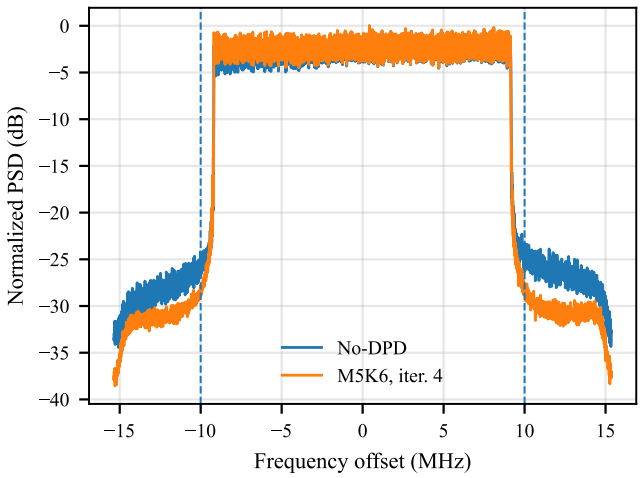


Fig. 5. Normalized observation-path power spectral density before and after DPD for the selected M5K6 realization.

TABLE IV
SELECTED-CASE END-TO-END LINEARIZATION METRICS AT THE NOMINAL OPERATING POINT. NMSE IS COMPUTED AGAINST THE COMMON DESIRED WAVEFORM AND IS NOT POWER INTERPOLATED.

Condition	EVM (%)	ACPR (dBc)	NMSE (dB)
No-DPD	6.47	−26.57	−14.77
DPD	2.65	−30.88	−24.05

Table IV complements the symbol-domain EVM and spectral ACPR results with the end-to-end aligned waveform NMSE. When both received waveforms are evaluated against the same original desired reference, NMSE improves from -14.77 to -24.05 dB, corresponding to a 9.28-dB reduction in normalized waveform-error power. As described in Section III-B, the acquisition-time input-referenced NMSE is not used here; all reported NMSE values are computed against the common desired waveform.

D. Repeatability and Statistical Stability

Repeatability was evaluated through 20 paired no-DPD and DPD hardware runs at $G_{\mathrm{TX}} = -22.25$ dB. Each pair employed a separately generated 5G NR waveform seed while retaining the same hardware gain distribution and the frozen M5K6 coefficient set from the local-range verification. All 20 measurements met the acceptance criteria.

The mean no-DPD and DPD EVM values were 6.458%

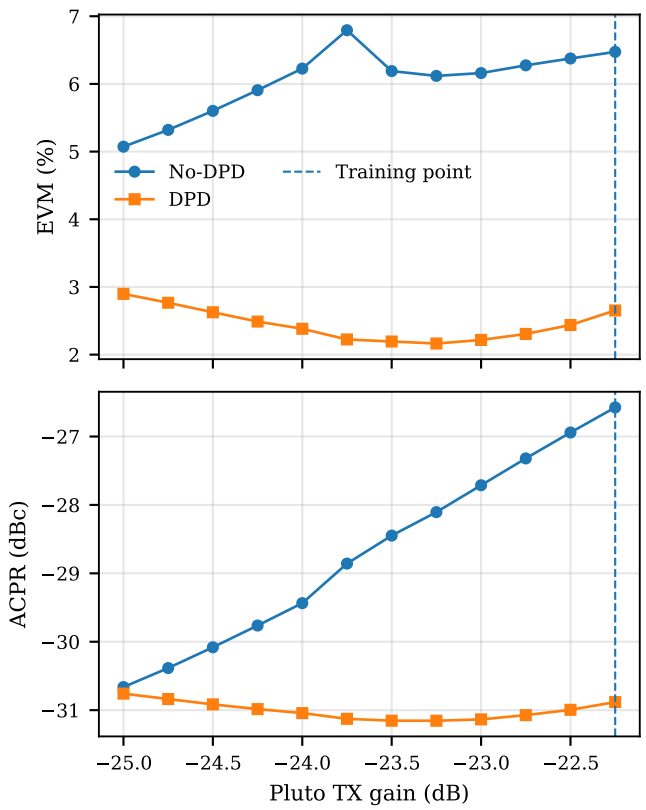


Fig. 6. Local back-off-side operating-range verification. A frozen M5K6 predistorter was held fixed while the Pluto transmit gain was swept from the nominal point toward lower drive levels in 0.25-dB increments.

TABLE V
REPEATABILITY OVER 20 PAIRED HARDWARE RUNS

Comparison	**ΔEVM mean ± std. (pp)**	**ΔACPR mean ± std. (dB)**
Same TX gain	3.765 ± 0.149	4.267 ± 0.319
Same RX power	2.536 ± 0.277	0.653 ± 0.304
Same detector power	3.241 ± 0.370	1.359 ± 0.373

and 2.693%, respectively, while mean ACPR improved from -26.756 to -31.023 dBc. The DPD waveform reduced mean Pluto-received power by 2.271 dB and detector-measured power by 2.232 dB, so the repeated-run analysis was also referenced to the no-DPD sweep at equivalent power.

For the detector-referenced comparison, the 95% confidence intervals were [3.068, 3.414] percentage points for EVM and [1.185, 1.534] dB for ACPR. The detector-referenced equal-power intervals exclude zero by a wide margin. These results confirm that the measured linearization benefit is repeatable across waveform realizations and is not attributable to a single favorable capture or solely to DPD-induced power reduction.

V. CONCLUSION

This work demonstrated end-to-end MP-DPD linearization of a cascaded 5 GHz IF-to-26 GHz RF transmitter through a downconverted observation signal. Stage-wise characterization verified approximately linear IF and observation paths and localized the dominant nonlinearity in the integrated upconversion stage.

Evaluations across models with 6 to 36 coefficients confirmed that increasing complexity did not yield monotonic improvement. A low-complexity 6-coefficient model (M1K3) achieved comparable EVM and ACPR performance to a 36-coefficient structure (M5K6), making it the preferred implementation.

Nominally, DPD reduced EVM from 6.47% to 2.65%, improved ACPR by over 4 dB, and enhanced end-to-end aligned waveform NMSE by 9.28 dB. Across 20 paired runs, detector-referenced equal-power improvements averaged 3.24 percentage points (EVM) and 1.36 dB (ACPR), while local sweeps demonstrated consistent back-off stability.

In summary, FR2 behavioral linearization meeting 256-QAM EVM targets is achievable with modest MP complexity and low-cost downconversion, provided that stage localization, observation-path linearity, and equal-power referencing are rigorously enforced. Current limitations include operation in moderate compression and the restriction to comparisons among MP structures of varying complexity; alternative topologies (e.g., neural networks) and power-interpolated NMSE were not benchmarked.

Future work will extend evaluation toward the measured 1-dB compression point using both fixed-coefficient and retrained DPD. Additional directions include adaptive step-size or validation-based stopping, thermal tracking, and extension to GMP and neural-network-based behavioral models [11].

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